TangiQuest Maze Game: RFID-Based Tangible HCI Game

An RFID-Driven Tangible Interface for Physical Maze Navigation

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Introduction

TangiQuest is an RFID-based Tangible User Interface (TUI) game built for the HCI course at AIUB, where players scan physical NFC cards to guide a robot through a 2-D maze. Following the Tangible Bits paradigm, keyboard/mouse input is fully replaced by four labeled NFC cards with strong affordance — each card's purpose is instantly understood without instruction.

An Arduino UNO reads card UIDs via an MFRC522 module and relays commands to a browser-based game through the Web Serial API. A green LED and beep confirm valid moves, while a red LED and error tone signal collisions, providing immediate multimodal feedback that reinforces the physical interaction.

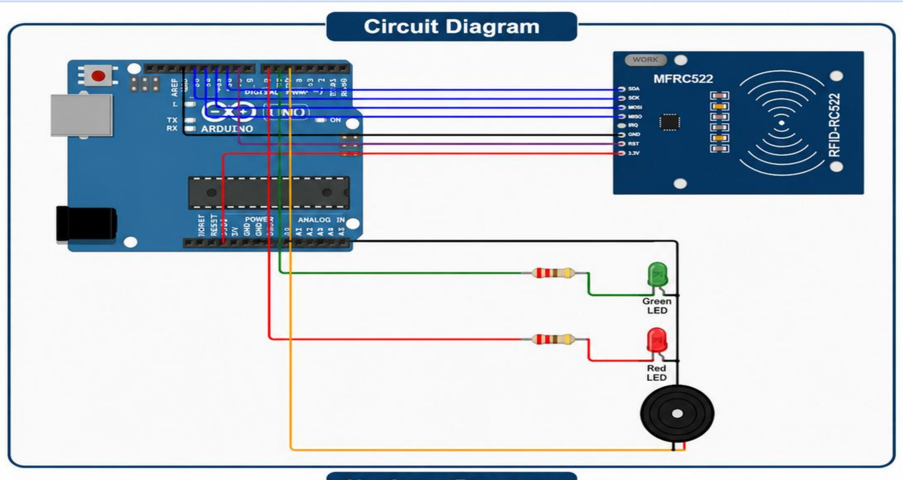
Methods — Hardware

This system uses an Arduino UNO paired with an RFID-RC522 module to translate physical card taps into serial game commands. Each of four NFC cards carries a unique UID mapped to a specific action, with immediate LED and buzzer feedback on every scan. The Arduino communicates with the browser-based game over USB serial at 9600 baud.

Components:

- Arduino UNO RFID-RC522 (SPI)
- 4 × NFC cards (UID-mapped)
- Green LED D4
- Red LED D5
- Piezo buzzer D6
- USB Type-B

Firmware Polls for a card each loop, reads the UID, maps it to a command (START / MOVE / LEFT / SUBMIT), sends it over Serial, and triggers GPIO feedback instantly.



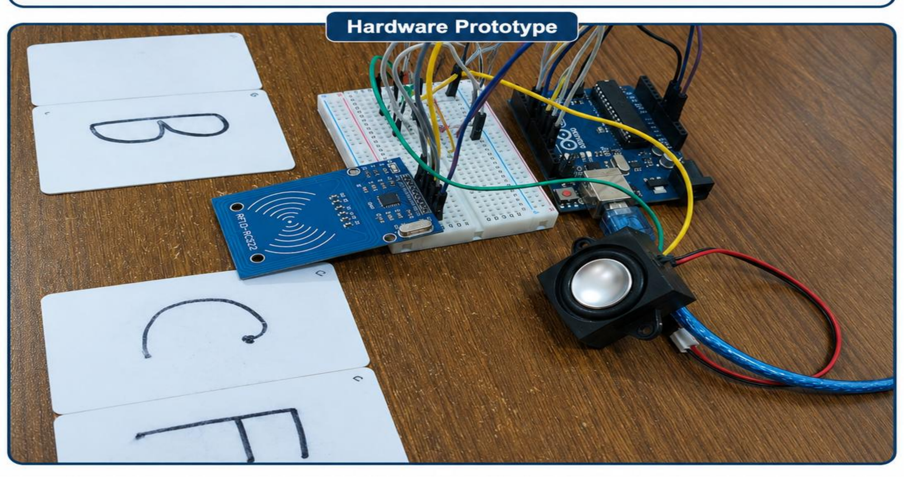


Fig. 1 — Hardware Simulation

System Architecture

Three-layer design: (1) Physical Layer — NFC cards trigger the RC522; (2) Communication Layer — Arduino relays commands over USB Serial via the Web Serial API; (3) Application Layer — the browser game engine processes commands and updates the grid state.

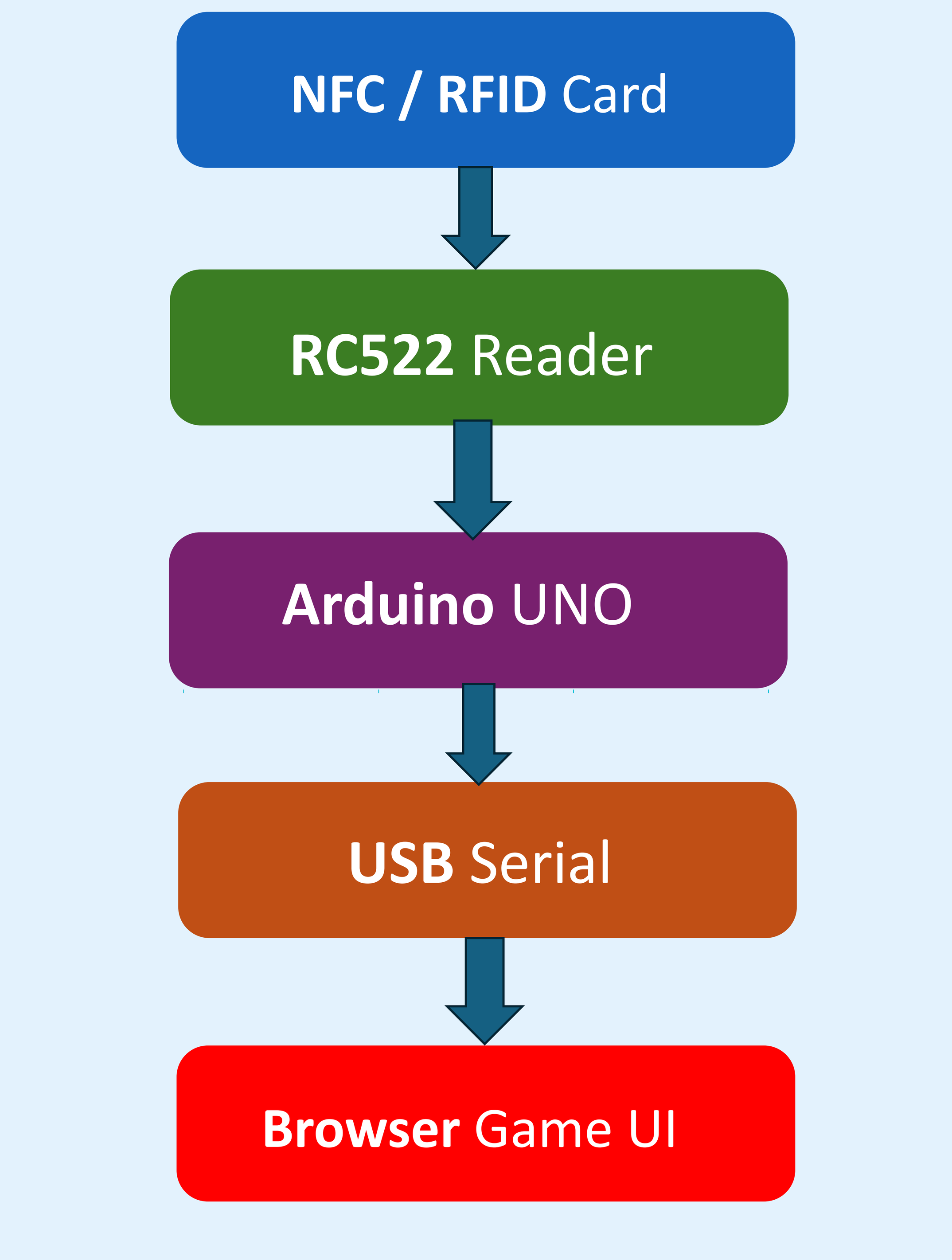


*Fig. 2 — Data-flow diagram: NFC Card → **RC522** → Arduino → USB Serial → Browser Game*

Methods — Software

The game runs as a browser-based web app (HTML, CSS, JavaScript) with no installation required. It connects to the Arduino via the **Web Serial API** at 9600 baud.

- **How it works:** Arduino sends a command string (e.g., COMMAND:MOVE) over USB Serial Browser parses the command and maps it to a game action: START, MOVE, LEFT, or SUBMIT The 6×6 grid re-renders instantly after every valid card scan
- **Level System:** 15 levels are stored as JavaScript config objects — each defines the robot start position, treasure location, wall tiles, and optimal move count. Difficulty increases by adding more wall tiles, not by changing the grid size.
- **Scoring:** Score = Base Points − (5 × extra moves used). Easy = 15 pts base, Medium = 35 pts, Hard = 50–75 pts. Exceeding 2× the optimal path yields zero points.
- **Audio:** 11 .wav files cover every game event (move, blocked, correct, level-ready, etc.) using the HTML5 Audio API.

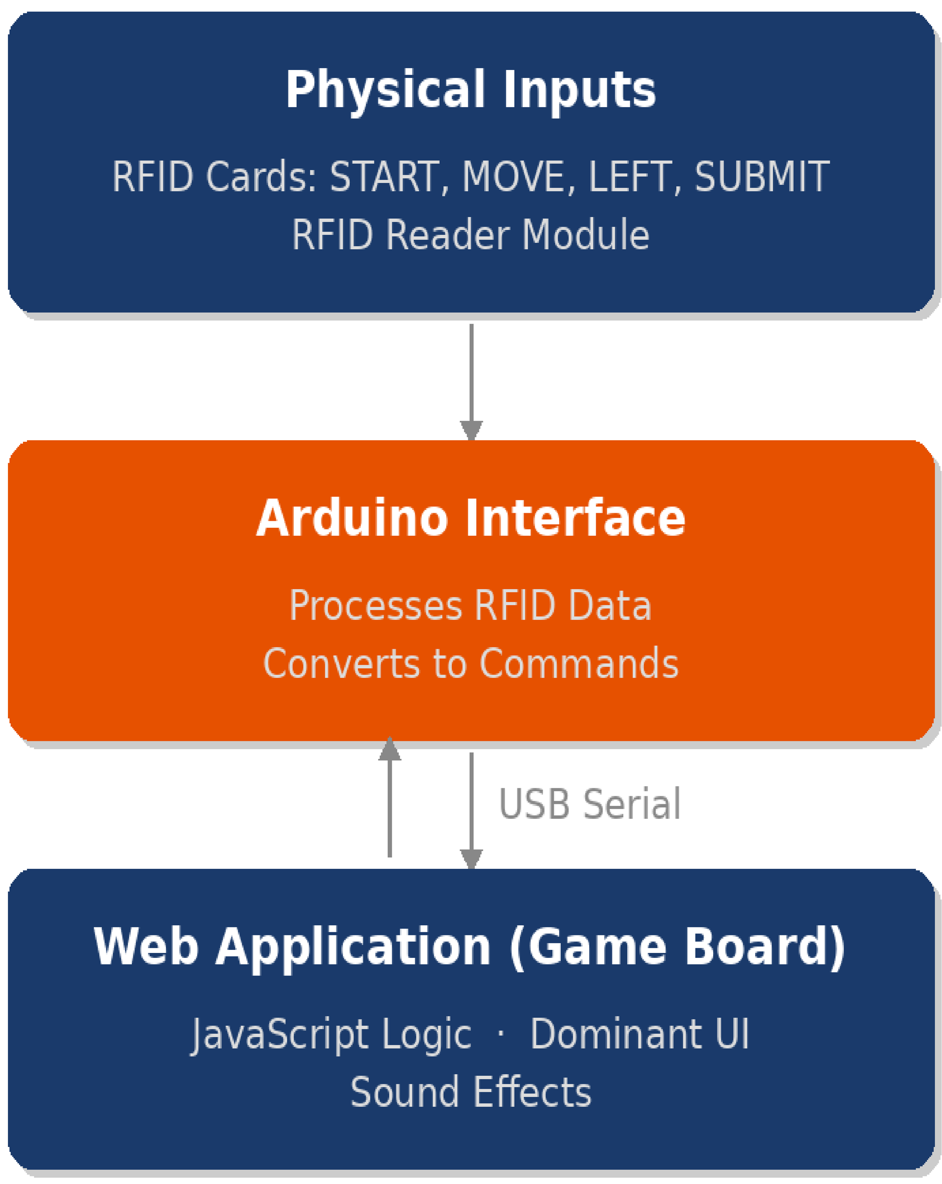


Fig. 3 — TangiQuest System Architecture

Results

All four NFC/RFID card commands (START, MOVE, LEFT, SUBMIT) were detected reliably. Card scan latency averaged below 100 ms with near-zero false positives across 500+ scans in test sessions. The Web Serial API connection remained stable for extended play sessions (30+ minutes).

Level Completion Summary

- **5 Easy Levels (L1–L5)**
- **5 Medium Levels (L6–L10)**
- **5 Hard Levels (L11–L15)**

Scoring Mechanism

Score = base points − (5 × excess moves used beyond the optimal path length). Easy base = 15 pts, Medium = 35 pts, Hard = 50–75 pts. Penalty encourages efficient pathfinding. Players score 0 if moves exceed 2× optimal length.

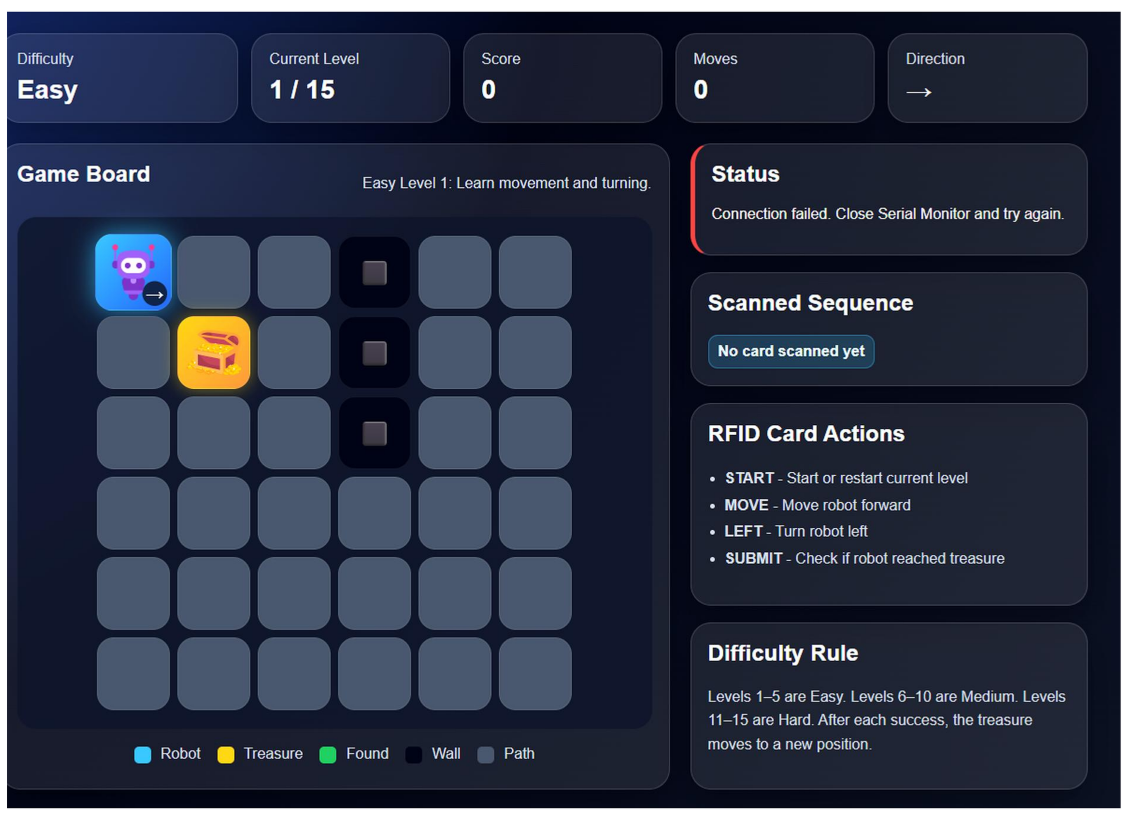


Fig. 4 — TangiQuest Maze Game Dashboard

NFC Card Command Reference

Card 1 — START

Reset level & return robot to start

Card 2 — MOVE

Claude responded: Move robot one step forward in current direction; wall hits are rejected

Card 3 — LEFT

Rotate robot 90° counter-clockwise

Card 4 — SUBMIT

Checks whether the robot currently occupies the treasure cell. If yes: level complete — score calculated and displayed. If no: error buzz.

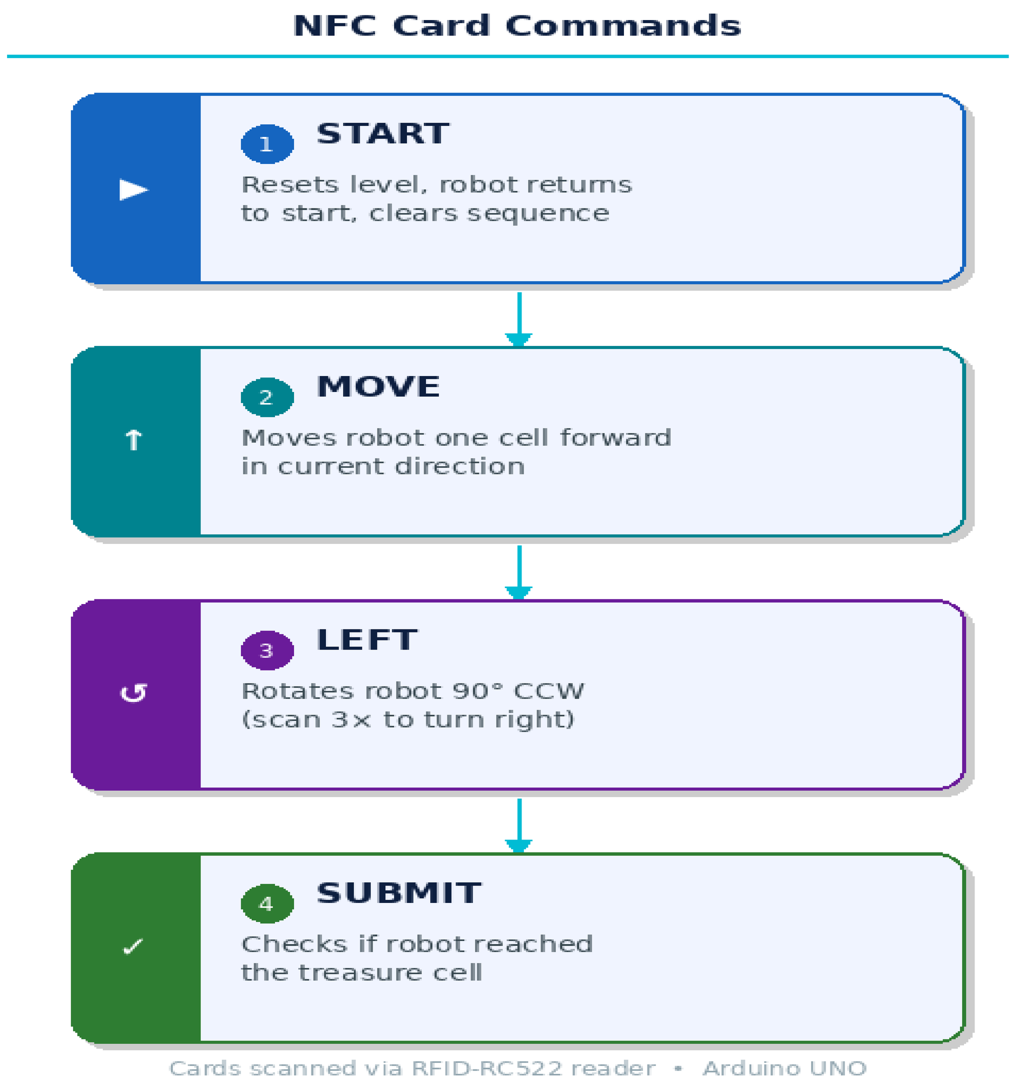


Fig. 5 — NFC Card Command Flow

Key Findings

- RFID scan latency <100 ms across 500+ scans
- Card interaction intuitive — no tutorial needed
- LED/buzzer feedback sped up error correction
- All 15 levels completable, no unsolvable configs
- Cards adopted faster than keyboard controls
- Score system effectively penalized inefficient paths

Conclusion & Future Work

Physical RFID cards make TangiQuest instantly playable — their strong affordances eliminate tutorials, LED/buzzer feedback confirms every action, and the large reader surface keeps inputs fast per Fitts' Law. All 15 levels scale difficulty through maze complexity alone, with no hardware changes needed.

Future Enhancements

- Dedicated RIGHT card for smoother directional control
- Cooperative multiplayer with dual RC522 readers
- Bluetooth LE for wireless, cable-free play
- Level editor for custom instructor-designed mazes
- Analytics dashboard for HCI research tracking

References

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